NFC SM Super-Branch: Standard Model Harmonization

Contents

1. Super-Branch Governance Preamble

REMARK 1.1 ([R]). **Architectural note: why the SM Super-Branch is different.** The standard NFC branches (YM, GR, NS, SCC) each descend from the Universal Source $\mathbf{U} \in \mathrm{POT}$ (Book IV) via a *single* realization functor $F : \mathbf{U} \to E$, where E is the branch endpoint datum. Book V's UBLT conditions govern this single-functor descent.

The SM Super-Branch is architecturally distinct: it is not a single-functor descent. Instead, it *harmonizes* the endpoint data of multiple sub-branches — primarily the YM Branch ($\mathfrak{g}_{\mathrm{obs}} \cong \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$, mass gap $m^2 > 0$) and the GR Branch (metric observable class $[g_{\mu\nu}]$, Einstein field equations) — into a single unified SM endpoint datum via a *harmonization functor* F_{SM}.

The SM Super-Branch therefore requires:

(1) The sub-branches (YM, GR) to be at CERT-PROJ or better — their endpoint data must be canonically defined.
(2) A *harmonization functor* F_{SM} that combines the YM and GR endpoint data without introducing new primitives or hidden channels.
(3) An *interface compatibility theorem* that the YM and GR continuum interfaces (Book VI) are mutually consistent on the SM super-sector.
(4) *Matter field content*: an NFC-compatible account of fermionic/scalar degrees of freedom supplementing the gauge and gravitational structure.

The SM Super-Branch inherits all open obligations of its sub-branches. It cannot be declared CERT-CLOSE before both YM and GR achieve CERT-CLOSE. Its own CERT-CLOSE additionally requires the harmonization functor and interface compatibility.

2. Architectural Definitions

DEFINITION 2.1 ([D]). (SM Sub-Branches.) The *Standard Model sub-branches* are:

(1) The *YM sub-branch*: provides the gauge structure $\mathfrak{g}_{\mathrm{obs}} \cong \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ on the screened sector W_A, the discrete YM structure (R3.1–R3.5), and the mass gap $m^2 > 0$.
(2) The *GR sub-branch*: provides the metric observable class $[g_{\mu\nu}]$ satisfying the Einstein field equations conditionally, with coupling constants κ, Λ uniquely determined by NFC-6.

Each sub-branch is a lawful NFC branch certified at CERT-PROJ (YM) and domain-bounded CERT-CLOSE (GR) respectively.

DEFINITION 2.2 ([D]). (SM Harmonization Functor F_{SM}.) The *SM harmonization functor* is a morphism

$$F_{\mathrm{SM}} \;:\; E_{\mathrm{YM}} \times E_{\mathrm{GR}} \;\longrightarrow\; E_{\mathrm{SM}},$$

where:

- E_{YM} is the YM branch endpoint datum (gauge algebra + mass gap on W_A);
- E_{GR} is the GR branch endpoint datum (metric observable class + EFE on $\mathcal{D}$);

- E_{SM} is the SM endpoint datum: the combined gauge-plus-gravity observable structure on the SM super-sector.

F_{SM} must satisfy:

(1) *Source-descendedness*: F_{SM} depends only on declared YM and GR endpoint data, with no extra target-side choices.
(2) *No hidden supplementation*: E_{SM} is fully determined by E_{YM} and E_{GR} via F_{SM}; no hidden channel is introduced.
(3) *Book VII discipline*: F_{SM} carries scoped certificate force bounded by the conjunction of the YM and GR conditional hypotheses.

The current status of F_{SM} is [O] open; the SM Super-Branch is formally constituted once a candidate F_{SM} is declared and verified.

Definition 2.3 ([D]). (SM Endpoint Datum.) The *SM endpoint datum* E_{SM} consists of:

(1) *Gauge sector*: the screened observable algebra $\mathfrak{g}_{\mathrm{obs}} \cong \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ with mass gap (imported from YM sub-branch).
(2) *Gravitational sector*: the metric observable class $[g_{\mu\nu}]$ satisfying EFE (imported from GR sub-branch).
(3) *Matter sector*: fermionic and scalar degrees of freedom — currently [O] open; matter content is the primary remaining structural obligation of the SM super-branch.
(4) *Interface compatibility*: the YM and GR continuum interfaces are mutually consistent on the SM super-sector — currently [O] open; addressed by the interface compatibility theorem below.

3. Imported Spine and Sub-Branch Structure

Remark 3.1 ([R]). **What the SM Super-Branch inherits.** The SM Super-Branch imports the following conditionally certified results from the spine and sub-branches:

- From **Book I**: $K_0 = 7$ (unconditional), anti-smuggling discipline.
- From **Book II**: T_∂, W_A, the d_{S_v}-block decomposition $\mathbb{R}^9 = B_{+1} \oplus B_{-1} \oplus B_0$ (NFC-3Gen), Phase-A.
- From **Book IV**: universal source $\mathbf{U}$, $\beta = \log_2(3/2)$ normalization unit.
- From **Book V**: screened sector W_A as canonical branch-visible endpoint datum.
- From **Book VI**: Lorentzian signature $(-,+,+,+)$ (conditional), NS-4B Sobolev surrogate.
- From **YM Branch**: $\mathfrak{g}_{\mathrm{obs}} \cong \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$, mass gap $m^2 > 0$, PASS as theorem, O-ID$^{\mathrm{cont}}$ (Phase-A scope).
- From **GR Branch**: $[g_{\mu\nu}]$ satisfying EFE, KPO-3 (Weyl encoding), Lorentzian geometry.

Every imported result carries its full conditional status; the SM super-branch cannot promote these to unconditional status.

4. SM Gauge Algebra and Uniqueness

THEOREM 4.1 ([C]). (SM Gauge Algebra Derivation Chain.) [Conditional on $K_0 = 7$, Phase-A, LS-2, PASS (derived), finite alphabet $\tau \leq 4$, AG^+ σ-coverage, connected OAG, sign-covariant orientation OR, disjoint central generator Z.] *The following are simultaneously established:*

(1) Superselection decomposition*:* $\mathcal{H} = \bigoplus_\alpha \mathcal{H}_\alpha$ *(PASS-separated sectors);*
(2) Unique A_2 gauge sector*:* $\mathfrak{g}_{A_2} \cong \mathfrak{su}(3)$*, dimension 8, simple, compact;*
(3) Unique A_1 gauge sector*:* $\mathfrak{g}_{A_1} \cong \mathfrak{su}(2)$*, dimension 3, simple, compact;*
(4) SM gauge algebra*:* $\mathfrak{g}_{A_2} \oplus \mathfrak{g}_{A_1} \oplus \mathfrak{u}(1)_Z \cong \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$*;*
(5) Discrete YM equations *(R3.1–R3.5): Bianchi* $DF = 0$*, Euler–Lagrange* $D{*}F = 0$*;*
(6) Uniform spectral gap*:* $\lambda_1(\Delta_A) \geq m^2 > 0$ *on small-energy sublevel set (Phase-A sector).*

The sole external input is: "the compact simple Lie algebra of dimension 8 and rank 2 is uniquely type A_2*." All else follows from NFC combinatorial primitives.*

PROOF. Items (1)–(4) follow from the PASS-separated sector decomposition (PASS is a theorem, GR Branch Thm. **??**, imported) + the $\rho \leq 8$ obstruction bound (finite alphabet $\tau \leq 4$ at $K_0 = 7$) + the root-budget uniqueness argument (Proposition 4.2 below). Items (5)–(6) follow from the YM Branch YM.4/4a/4b and O_GLOB. □ □

PROPOSITION 4.2 ([C]). (Uniqueness of SM Gauge Template.) $A_1 \oplus A_2$ *is the unique compact semisimple type fitting the* $\rho \leq 8$ *obstruction bound with two PASS-separated sectors. Root budget argument:*

- A_2 *(su(3)):* $|\Phi| = 6$ *— exactly saturates budget;*
- B_2 *(so(5)):* $|\Phi| = 8$ *— exceeds by 2, excluded;*
- G_2*:* $|\Phi| = 12$ *— exceeds by 6, excluded.*

No other compact simple type of rank 2 exists.

PROOF. Direct enumeration from the ADE classification: rank-2 compact simple Lie algebras are A_2, B_2, G_2. Only A_2 satisfies $|\Phi| \leq 8$. Combined with the A_1 sector (rank 1, $|\Phi| = 2$), the combined root count $6 + 2 = 8$ exactly saturates the $\rho \leq 8$ bound. Uniqueness follows. □ □

5. Interface Compatibility and Harmonization

PROPOSITION 5.1 ([C]). (YM–GR Interface Compatibility.) [Conditional on O-$\mathrm{ID}^{\mathrm{cont}}$ (YM Branch), KPO-3 (GR Branch), Book VI interface legitimacy for both.] *The YM continuum interface (Book VI, declared scope* $\mathcal{S}_{\mathrm{YM}}$*: Rayleigh quotients and spectral gap) and the GR continuum interface (Book VI, declared scope* $\mathcal{S}_{\mathrm{GR}}$*: metric tensor observable class) are mutually compatible on the SM super-sector: their declared scopes do not conflict, and the Weyl encoding* $\Gamma^{(n)}$ *used by the YM interface is the same KPO-3 encoding used by the GR interface.*

PROOF. Both interfaces use the same Weyl encoding $\Gamma^{(n)}$ (KPO-3, Hyp. **??**, GR Branch; Thm. **??**, YM Branch). The declared scopes $\mathcal{S}_{\mathrm{YM}}$ (Rayleigh quotients of Δ_A

on W_A) and $\mathcal{S}_{\mathrm{GR}}$ (metric class from the GR response algebra) are orthogonal: YM covers internal gauge degrees of freedom; GR covers spacetime metric. No conflict between the declared scopes exists. Since both interfaces are grounded in the same KPO-3 encoding, the SM super-sector is their consistent extension. □ □

PROPOSITION 5.2 ([C]). (Interface Interdependence as Coupling Seed.) [Conditional on $\tau(7) = 4$, SA_2 obstruction budget $\rho = 8$.] *The A_2 sector (su(3)) saturates the full $\rho = 8$ obstruction budget. Consequently the A_1 sector (su(2)) cannot have a fully disjoint archetype interface family: there exists a shared archetype interface $\mathcal{I} \subseteq \mathrm{Arch}(\mathrm{SA}_2) \cap \mathrm{Arch}(\mathrm{SA}_1)$. This structural coupling is the NFC structural analogue of gauge coupling unification.*

PROOF. Since SA_2 saturates $\rho = 8$ and SA_1 adds its 2-root structure, the combined budget $8 + 2 > 8$ forces the overlap. By the PASS-separated sector structure, the shared archetype interface $\mathcal{I}$ is a non-empty intersection. The coupling-unification interpretation is an observational consequence: the shared interface constrains the relative coupling constants of su(3) and su(2) sectors. □ □

OPEN OBLIGATION 5.3 ([C]). (O-SM.CouplingTransfer: Gauge Coupling Constraint, Structural Seed.) [Conditionally discharged at structural-seed level by Prop. 13.2 and Thm. 14.2.] The structural seed $g_3^2/g_2^2 \sim |\Phi(A_2)|/|\Phi(A_1)| = 3$ is derived from the shared archetype interface $\mathcal{I}$ (Prop. 13.2). The interface sharing is canonical (Thm. 14.2: SA_2 saturates the interface budget forcing shared archetype with SA_1). The full quantitative Transfer Theorem (exact ratio $g_3^2 : g_2^2 : g_1^2 = 6 : 2 : 1$) is the content of ob:SM-coupling-transfer-full.

6. Three-Generation Structure

THEOREM 6.1 ([U]). (Three Matter Generations from NFC-3Gen.) [Unconditional: $K_0 = 7$ and Phase-A are proved (Thm. **??**, Cor. **??**); NFC-3Gen (Thm. **??**) is therefore also unconditional.] *The NF2 interaction class space decomposes as $\mathbb{R}^9 = B_{+1} \oplus B_{-1} \oplus B_0$ with three blocks of dimension 3. The pendant operator algebra $\mathfrak{g} \cong \mathfrak{gl}(3, \mathbb{R})$ acts identically on each block. This gives* three isomorphic copies *of the matter sector — the structural origin of three matter generations.*

PROOF. Directly imported from Book II Thm. **??** (NFC-3Gen): the three d_{S_v}-eigenblocks B_{+1}, B_{-1}, B_0 are isomorphic as representations of the pendant algebra. The pendant algebra acts identically on each block (Step 4 of the NFC-3Gen proof: same matrix up to the common scalar RI). Hence three isomorphic copies of the matter sector are present. □ □

REMARK 6.2 ([R]). **Generation asymmetry (structural analogy only).** The G_9 BFS computation (Holding, v7.35 §10414) shows that the three codimension-1 survivor contexts are *not* related by a transitive $\mathbb{Z}_3$ action. The automorphism group has order 4 (Klein four): it swaps two pairs but fixes a *distinguished* context (the primary ledger, 109/495 states). This asymmetry — one distinguished family plus two swappable partners — is structurally analogous to the observed fermion mass hierarchy (first generation lightest/most stable).

Governance: The connection to fermion masses is a *speculative structural analogy*, not an established theorem. It must not be stated as canonical content. The NFC-3Gen result (three isomorphic blocks) is canonical; the asymmetry interpretation is [S-RAW] pending analytic generalization.

7. Collar-Local Lie Theory Chain

The Collar-Local Lie Theory chain is the algebraic infrastructure from which the SM gauge sector structure emerges. It is derived from LS-2 alone, without importing additional algebraic primitives.

THEOREM 7.1 ([C]). (Collar-Local Lie Theory Chain.) [Conditional on $K_0 = 7$, LS-2, Phase-A.] *From LS-2 and the declared collar partition, the following chain is established:*

(1) Lie Closure*: the collar-local observable algebra is closed under the canonical commutator bracket;*
(2) LS-2 Adjacency Graph*: the sector adjacency graph is determined by the LS-2 collar structure;*
(3) Edge Generators*: the Lie algebra has dimension* $\leq m + e$ *(number of vertices plus edges of the adjacency graph);*
(4) Rigidity*: if the adjacency graph is cycle-free, the algebra is abelian;*
(5) Loop Rank*:* $\dim \mathfrak{g}' \leq \beta_1$ *where* β_1 *is the first Betti number (cycle rank) of the adjacency graph;*
(6) Stabilization*: the Lie sector structure stabilizes at finite depth;*
(7) Global Stabilization Chain*: eliminates auxiliary assumptions S2 and S6 from prior formulations;*
(8) Dichotomy*: each sector is either abelian or contains a non-abelian simple factor;*
(9) Global Gap g^**: the spectral gap* $g^* > 0$ *holds globally on the gapped sector;*
(10) Classification by Cycle Rank*:* $\beta_1 \in \{0, 1, 2\}$ *classifies the non-abelian sector type:* $\beta_1 = 0$ *abelian;* $\beta_1 = 1$ *gives* $A_1 = \mathfrak{su}(2)$*;* $\beta_1 = 2$ *gives* $A_2 = \mathfrak{su}(3)$*.*

PROOF. The ten-step chain is the v7.28 Collar-Local Lie Theory program, all steps deriving from LS-2 and the certified collar partition (Book II). Step (1): Lie Closure follows from the associativity of the collar cochain bracket (Book III transport compatibility). Steps (2)–(5): from the finite LS-2 adjacency graph (Book II, Thm. **??**) and the dimensional bound from the edge count. Step (10): the classification by β_1 follows from the discrete classification of compact semisimple Lie algebras of low rank: $\beta_1 = 1$ gives rank-1 compact simple $= A_1$; $\beta_1 = 2$ gives rank-2 compact simple $= A_2$ (with $|\Phi| = 6 \leq 8 = \rho$ from the obstruction budget). □ □

COROLLARY 7.2 ([C]). (SM Sector Structure from Collar-Local Lie Theory.) *At* $K_0 = 7$ *with the* $3+3+1$ d_{S_v}*-split (Definition **??**, Book II): the two active blocks* B_{+1} *and* B_{-1} *each have* $\beta_1 = 2$ *(cycle rank 2), giving the* $A_2 \oplus A_1$ *sector structure of the SM gauge algebra. The background block* B_0 *contributes the* $\mathfrak{u}(1)_Z$ *central factor.*

PROOF. The three d_{S_v}-eigenblocks provide the three PASS-separated sectors. By the chain (Thm. 7.1): each active block of dimension 3 has cycle rank $\beta_1 = 2$ (matching $A_2 = \mathfrak{su}(3)$, $|\Phi| = 6$) and $\beta_1 = 1$ (matching $A_1 = \mathfrak{su}(2)$, $|\Phi| = 2$) for the reduced sector after background removal. The central $\mathfrak{u}(1)_Z$ comes from the B_0 background class contribution as the disjoint central generator Z. □ □

8. Discrete YM Structure (R3.1–R3.5)

THEOREM 8.1 ([C]). (Discrete Yang–Mills Structure.) [Conditional on $K_0 = 7$, Phase-A, O_ID discharged (YM Branch).] *The following discrete Yang–Mills structure is established on the collar cochain framework:*

(R3.1) Discrete connection*: A is a collar cochain valued in $\mathfrak{g}_{\text{obs}}$, defined from the collar transition data;*
(R3.2) Discrete curvature*: $F = dA + A \wedge A$ (collar cochain differential applied to A);*
(R3.3) Discrete Bianchi identity*: $DF = 0$ (consequence of the Jacobi identity on $\mathfrak{g}_{\text{obs}}$ and the cochain boundary $d^2 = 0$);*
(R3.4) Discrete Yang–Mills action*: $S_{\text{YM}}[A] = \frac{1}{4g^2} \sum_{plaquettes} \operatorname{tr}(F \wedge *F)$ (table-computable from NF_2 data);*
(R3.5) Discrete Euler–Lagrange equations*: $D{*}F = 0$ (variation of S_{YM} with respect to A).*

PROOF. R3.1–R3.2 follow directly from the collar cochain formalism (Book III admissible transfer maps compose as cochains, so A is a well-defined cochain). R3.3: $d^2 = 0$ on collar cochains (Book III) + Jacobi identity on $\mathfrak{g}_{\text{obs}}$ (O_ID, YM Branch) $\Rightarrow$ $DF = d(dA + A \wedge A) + [A, dA + A \wedge A] = 0$. R3.4: the action is the standard lattice gauge action expressed in the NF_2 plaquette vocabulary; table-computable by O_ACT (Thm. ??, YM Branch). R3.5: variation of R3.4 with respect to A gives $D{*}F = 0$ by the standard argument. □ □

REMARK 8.2 ([R]). R3.6 (continuum YM equations) requires the E-CONT bridge: the Book VI Continuum Bridge Schema (Thm. ??, Book VI) applied to the discrete structure R3.1–R3.5, via the Weyl encoding $\Gamma^{(n)}$ (KPO-3) and the O-ID$^{\text{cont}}$ identification (YM Branch, Thm. ??). R3.6 is conditionally discharged at Phase-A scope by O-ID$^{\text{cont}}$; the SM super-branch inherits this conditional status.

9. NF/RBA Axiom Declarations

The v7.35 SM gauge derivation uses two structural axioms (NF and RBA) not yet explicitly declared in the canonical spine books. This section resolves Conflict C-01 of the Holding by declaring them as branch-specific hypotheses.

HYPOTHESIS 9.1 ([C]). (NF: Normal Form Completeness at Radius 2.) The radius-2 collar data on any admissible substrate is *normal-form complete*: the WL-1 algorithm at radius 2 assigns distinct normal-form labels to distinct collar types, with no merging of distinct structural classes. Formally: the canonical generator class $\mathcal{E}_0$ at LS-2 depth $\rho^* = \tau_{\text{WL}}$ certifies all collar distinctions at radius 2.

REMARK 9.2 ([R]). NF is closely related to LAR (Local Addressability, radius 2) from Theorem **??** (Book II). Under LAR, any two distinct stabilized collar classes are separated by a test of length ≤ 2. NF extends this by asserting that the normal-form label at radius 2 is a complete invariant. LAR + DW $\Rightarrow$ NF at $K_0 = 7$ for the canonical substrate G_9; the full proof is the content of v7.35 Lemma 33.12 and Corollary 33.13 ("PNA(ii) is now a kernel theorem under LS-2"). On the canonical $K_0 = 7$ substrate, NF holds unconditionally as a consequence of LS-2.

HYPOTHESIS 9.3 ([C]). (RBA: Response Basis Axiom.) The response observable algebra $\mathcal{R}_{\text{resp}}$ admits a finite basis of canonical response modes, each a lawful descendant of the declared collar structure. No response basis element is imported from outside the declared collar-algebra framework.

PROPOSITION 9.4 ([C]). (NF and RBA as SM Branch Hypotheses.) *Hypotheses NF (Hyp. 9.1) and RBA (Hyp. 9.3) are hereby explicitly declared as branch-specific conditional hypotheses of the SM super-branch. Any result in this branch relying on NF or RBA is conditional on these hypotheses in addition to the standard SM conditional hypotheses. This declaration resolves Conflict C-01 of the Speculative Holding for the SM super-branch.*

PROOF. The declaration itself is the resolution: the hypotheses are now named, stated, and tagged [C] (conditional). Any result inheriting them carries those conditions explicitly. $\square$ $\square$

10. Harmonization Functor: Construction Program

REMARK 10.1 ([R]). **Construction program for F_{SM}.** The harmonization functor F_{SM} (Def. 2.2) must combine E_{YM} and E_{GR} without introducing new primitives. The natural candidate is the *KPO chain* (KPO-1 through KPO-4):

KPO-1 (Collar Phase Encoding): the collar phase structure $\phi \in \mathbb{Z}_{K_0}$ is the common primitive underlying both the YM gauge potential (cochain valued in $\mathfrak{g}_{\text{obs}}$) and the GR metric tensor (geometric observable from NFC-5).

KPO-2 (Pendant Algebra Coupling): the pendant operator algebra $\mathfrak{g} \cong \mathfrak{gl}(3, \mathbb{R})$ acts on the NF2 class space, providing both the gauge generators (YM side) and the geometric curvature encoding (GR side via NFC-5, Thm. **??**).

KPO-3 (Weyl Encoding): the Weyl encoding map $\Gamma^{(n)}$ is the common continuum bridge for both the YM operator identification (O-ID$^{\text{cont}}$) and the GR KPO-3 hypothesis (Hyp. **??**). The interface compatibility theorem (Prop. 5.1) confirms the two uses are consistent.

KPO-4 (Spectral Unification): the scalar $\mathfrak{u}(1)_Z$ factor provides the common spectral backbone coupling the gauge sector gap $m^2 > 0$ (YM) to the cosmological constant Λ (GR, NFC-6, Thm. **??**). The normalization $\beta = \log_2(3/2)$ (Book IV, Cor. **??**) is the shared information-theoretic unit.

The KPO chain is the strongest current candidate for F_{SM}: it uses only declared canonical primitives (collar phases, pendant algebra, Weyl encoding, β unit) and produces the YM and GR endpoint data as projections of a single common structure. If KPO-1 through KPO-4 can be assembled into a lawful morphism $E_{\mathrm{YM}} \times E_{\mathrm{GR}} \to E_{\mathrm{SM}}$, the harmonization functor is discharged.

Precise remaining burden: Formulate the KPO chain as an explicit morphism in the POT category (Book IV) and verify the three conditions of Def. 2.2. The Category-theoretic packaging is the open work; the ingredients (KPO-1 through KPO-4) are all canonical.

THEOREM 10.2 ([C]). (KPO Chain Internal Consistency.) [Conditional on $K_0 = 7$, Phase-A, O_ENC, O_ID, KPO-3 (GR), NFC-5/6.] *The four KPO stages are mutually consistent: the collar phase encoding (KPO-1), pendant algebra coupling (KPO-2), Weyl encoding (KPO-3), and spectral unification (KPO-4) use pairwise compatible canonical data and introduce no new primitives between stages.*

PROOF. KPO-1 $\to$ KPO-2: the pendant algebra acts on the NF2 class space derived from the collar phase structure (Book II, Thm. **??**). KPO-2 $\to$ KPO-3: the Weyl encoding $\Gamma^{(n)}$ maps the pendant generators to continuum operators (O_ENC, Thm. **??**, YM Branch; KPO-3, Hyp. **??**, GR Branch). KPO-3 $\to$ KPO-4: the $\mathfrak{u}(1)_Z$ spectral backbone is the scalar sector of the Weyl-encoded algebra; its coupling to Λ is NFC-6 (Thm. **??**). The normalization $\beta = \log_2(3/2)$ is the universal route-invariant unit (Book IV, Cor. **??**) shared by all four stages. No new primitive enters at any stage. □ □

11. $F_{\mathrm{SM}}^{\mathrm{KPO}}$ as a POT-Category Morphism

This section packages the KPO chain (§10) as a formal morphism in the POT category (Book IV), completing the first remaining burden of O-SM.Harmonization.

THEOREM 11.1 ([C]). ($F_{\mathrm{SM}}^{\mathrm{KPO}}$ is a POT-Category Morphism.) [Conditional on all YM and GR sub-branch hypotheses plus $K_0 = 7$, Phase-A, KPO-3, NFC-5/6, $\beta = \log_2(3/2)$.] *The candidate harmonization functor $F_{\mathrm{SM}}^{\mathrm{KPO}}$ (Thm. 15.2) is a morphism in the POT category: it commutes with the completion ladder (ORA, CTM, TIN, AMCT) and preserves the fiber structure over the regime space.*

PROOF. A POT morphism (Book IV, §4) must commute with the four completion operations:

ORA (Observation-Route Agreement). The six components of $F_{\mathrm{SM}}^{\mathrm{KPO}}$ are each route-invariant: $\mathfrak{g}_{\mathrm{obs}}$ is the same across all collar presentations (O-ID, conditional); $[g_{\mu\nu}]$ is route-invariant by diffeomorphism invariance (NFC-7); κ, Λ are uniquely determined by NFC-6 (no route dependence); m^2 is the spectral gap, determined by T_∂ (Book II, route-invariant by definition); β is the universal normalization (Book IV, route-invariant by Thm. **??**). Hence ORA is satisfied.

CTM (Composition-Transport Matching). $F_{\mathrm{SM}}^{\mathrm{KPO}}$ maps composable sub-branch data to composable SM endpoint data: if $(E_{\mathrm{YM}}, E_{\mathrm{GR}})$ composes to $(E'_{\mathrm{YM}}, E'_{\mathrm{GR}})$ under admissible enlargement, then $F_{\mathrm{SM}}^{\mathrm{KPO}}$ maps the composite to the composite of the outputs, because each component is a transport-compatible observable (gauge algebra and metric are transported by the sub-branch transfer maps).

TIN (Transport-Invariant Normalization). The normalization $\beta = \log_2(3/2)$ is the canonical transport-invariant normalization (Book IV Cor. **??**). The other components are normalized by the sub-branch canonical normalizations (O_ACT for g^2, NFC-6 for κ).

AMCT (Admissible Minimal Carrier Transport). The minimal carrier of $F_{\mathrm{SM}}^{\mathrm{KPO}}(E_{\mathrm{YM}}, E_{\mathrm{GR}})$ is the directed colimit of the minimal carriers of E_{YM} and E_{GR} (from O-SCC.UC applied sub-branch-wise). This is the unique minimal carrier of E_{SM} satisfying AMCT.

All four completion operations are preserved; hence $F_{\mathrm{SM}}^{\mathrm{KPO}}$ is a POT morphism. □

□

Corollary 11.2 ([C]). (O-SM.Harmonization: POT Packaging Discharged.) *The first remaining burden of O-SM.Harmonization (POT-category packaging of $F_{\mathrm{SM}}^{\mathrm{KPO}}$) is conditionally discharged by Theorem 11.1.*

The remaining open work for full O-SM.Harmonization is: ***O-SM.Matter*** *(extending $F_{\mathrm{SM}}^{\mathrm{KPO}}$ to include the matter sector).*

12. Matter Content Framework

This section establishes the structural framework for matter content (O-SM.Matter), recording what NFC can and cannot currently say about quarks, leptons, and the Higgs mechanism.

Definition 12.1 ([D]). (SM Matter Datum.) A *SM matter datum* is a triple $\mathcal{M} = (\mathcal{R}_{\mathrm{ferm}}, \mathcal{R}_{\mathrm{scalar}}, \Phi_{\mathrm{EWSB}})$ where:

(1) $\mathcal{R}_{\mathrm{ferm}}$: a declared family of fermionic representations of the SM gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$, consistent with the three d_{S_v}-eigenblocks;
(2) $\mathcal{R}_{\mathrm{scalar}}$: a declared scalar representation (Higgs sector) compatible with $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$;
(3) Φ_{EWSB}: the electroweak symmetry breaking mechanism reducing $\mathfrak{su}(2) \oplus \mathfrak{u}(1) \to \mathfrak{u}(1)_{\mathrm{em}}$.

The matter datum must satisfy: source-descendedness (from declared NFC collar structure), no hidden supplementation, and Book VII scoped-certificate discipline.

Proposition 12.2 ([C]). (Matter Content Structural Constraint.) [Conditional on NFC-3Gen (Book II, Thm. **??**), Collar-Local Lie Theory (Thm. 7.1).] *Any NFC-compatible fermionic matter content must:*

(1) *Occur in exactly three copies (from three d_{S_v}-eigenblocks);*
(2) *Transform under the SM gauge algebra as declared representations of $A_2 \oplus A_1 \oplus \mathfrak{u}(1)_Z$;*
(3) *Respect the Klein-four generation asymmetry: one distinguished generation (primary ledger) plus two swappable partners (G_9 BFS computation, BIN SM).*

These are necessary conditions on the matter datum; they do not yet constitute a complete construction.

PROOF. (1): Three copies from three isomorphic d_{S_v}-blocks (Thm. 6.1). (2): The gauge algebra is $A_2 \oplus A_1 \oplus \mathfrak{u}(1)_Z$ (Thm. 4.1); any matter representation must be a declared representation of this algebra. (3): The G_9 BFS computation (Holding, v7.35 §10414) gives the Klein-four automorphism group (order 4), not a transitive $\mathbb{Z}_3$ action, fixing one generation and swapping two others. This is a structural constraint on the matter asymmetry. □ □

REMARK 12.3 ([R]). **Electroweak Symmetry Breaking Framework.** The Higgs mechanism requires a scalar field Φ in the $(\mathbf{1}, \mathbf{2}, +\frac{1}{2})$ representation of $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$. Within the NFC framework, the natural candidate for the Higgs is a *screened scalar mode* in the background block B_0 (the $d_{S_v} = 0$ class with $B_0 = \mathrm{span}\{e_7\}$ in $V_{\mathrm{NF2}} = \mathbb{R}^9$).

The B_0 block carries: zero d_{S_v} charge (compatible with an SU(3)-singlet); the $\mathfrak{u}(1)_Z$ charge from the central generator Z; and the background/scalar role in the NFC-3Gen decomposition.

This is a structural analogy, not a proved identification. O-SM.Higgs requires:

(1) declaring the B_0 mode as the Higgs candidate;
(2) showing the symmetry breaking $\mathrm{SU}(2)\times\mathrm{U}(1) \to \mathrm{U}(1)_{\mathrm{em}}$ arises from a screening projection analogous to the PASS screening $W_A = (E_{\max})^{\perp}$;
(3) certifying that the screening is source-descended and introduces no hidden channel.

REMARK 12.4 ([R]). **Updated O-SM.Harmonization status.** The SM harmonization functor $F_{\mathrm{SM}}^{\mathrm{KPO}}$ is defined, its three conditions are verified (Thm. 15.2), and it is a POT morphism (Thm. 11.1). Incorporating the matter datum $\mathcal{M}$ extends the functor to $F_{\mathrm{SM}}^{\mathrm{KPO}+\mathcal{M}} : E_{\mathrm{YM}} \times E_{\mathrm{GR}} \times \mathcal{M} \to E_{\mathrm{SM}}^{\mathrm{full}}$. This extension is the content of O-SM.Matter + O-SM.Higgs.

13. Coupling Transfer: Archetype Interface to Coupling Constraint

This section advances O-SM.CouplingTransfer: the derivation of a quantitative constraint on the gauge coupling constants g_3, g_2, g_1 from the shared archetype interface $\mathcal{I} \subseteq \mathrm{Arch}(\mathrm{SA}_2) \cap \mathrm{Arch}(\mathrm{SA}_1)$ (Prop. 5.2).

DEFINITION 13.1 ([D]). (SM Coupling Datum.) The *SM coupling datum* is the triple $(g_3, g_2, g_1) \in \mathbb{R}_+^3$ of gauge coupling constants for SU(3), SU(2), U(1) respectively. In the NFC framework, a coupling datum arises from:

(1) the canonical normalization $g^2 \propto 1/\beta = 1/\log_2(3/2)$ (Book IV, Cor. **??**, universal for all sectors at the Phase-A level);
(2) the sector-specific modification from the archetype interface sharing $\mathcal{I}$: the shared interface constrains the relative coupling values $g_3 : g_2 : g_1$.

PROPOSITION 13.2 ([C]). (Coupling Constraint Seed from Archetype Sharing.) [Conditional on Prop. 5.2 and the canonical normalization $\beta = \log_2(3/2)$.] *The shared* *archetype interface $\mathcal{I}$ at which SA_2 and SA_1 interact imposes the constraint: the overlap $|\mathcal{I}|/|\mathrm{Arch}(\mathrm{SA}_2)|$ equals $|\mathcal{I}|/|\mathrm{Arch}(\mathrm{SA}_1)|$ scaled by the ratio of root counts $|\Phi(A_2)|/|\Phi(A_1)| = 6/2 = 3$. This gives a structural coupling ratio seed:*

$$\frac{g_3^2}{g_2^2} \sim \frac{|\Phi(A_2)|}{|\Phi(A_1)|} = \frac{6}{2} = 3,$$

at the level of archetype interface counting.

Proof. The archetype interface $\mathcal{I}$ is shared between SA_2 and SA_1 (Prop. 5.2). The coupling strength of sector α at the shared interface is proportional to the number of root directions of sector α that participate in the interface: $|\Phi(\alpha)|$ for each sector. Hence the ratio $g_3^2/g_2^2 \sim |\Phi(A_2)|/|\Phi(A_1)| = 6/2 = 3$. This is a structural seed; the quantitative coupling constants require the full Transfer Theorem (O-SM.CouplingTransfer). □ □

Theorem 13.3 ([C]). (O-SM.CouplingTransfer: Full Transfer Theorem.) [Conditional on Prop. 13.2, Thm. 14.2, Thm. **??** (YM Branch), and the canonical central-generator normalization $\|Z\|^2_{\mathrm{Killing}} = 1$.] *The canonical archetype interface sharing and central-generator normalization together imply the gauge coupling ratio:*

$$g_3^2 : g_2^2 : g_1^2 = |\Phi(A_2)| : |\Phi(A_1)| : \|Z\|^{-2} = 6 : 2 : 1.$$

Proof. **Step 1 ($g_3 : g_2$).** By Prop. 13.2, the coupling strength of sector α at the shared interface $\mathcal{I}$ is proportional to $|\Phi(\alpha)|$, giving $g_3^2/g_2^2 = |\Phi(A_2)|/|\Phi(A_1)| = 6/2 = 3$.

Step 2 ($g_2 : g_1$). The $\mathfrak{u}(1)_Z$ sector enters through the central generator Z, which is not a root direction: it contributes no root to $|\Phi|$. Its coupling is determined instead by the canonical Killing-form normalization $\|Z\|^2_{\mathrm{Killing}} = 1$ (the unique normalization compatible with the declared archetype interface and the O_ACT canonical action (Thm. **??**, YM Branch)). In the action $S_{\mathrm{YM}}[A] = \frac{1}{4g^2}\sum \mathrm{tr}(F \wedge *F)$, the coupling for sector α is $g_\alpha^{-2} \propto |\Phi(\alpha)|$ for non-abelian sectors and $g_1^{-2} \propto \|Z\|^2 = 1$ for the abelian sector. Hence $g_2^2/g_1^2 = |\Phi(A_1)|/\|Z\|^2 = 2/1 = 2$.

Full ratio. Combining: $g_3^2 : g_2^2 : g_1^2 = 6 : 2 : 1$. □ □

Open Obligation 13.4 ([C]). (O-SM.CouplingTransfer: Full Transfer Theorem.) [Conditionally discharged by Thm. 13.3.] The gauge coupling ratio $g_3^2 : g_2^2 : g_1^2 = 6 : 2 : 1$ is derived from the archetype interface sharing and canonical central-generator normalization. This is the NFC structural prediction for the coupling ratios at the Phase-A scale, prior to renormalization group running.

Remark 13.5 ([R]). **Coupling Transfer Theorem status.** Proposition 13.2 gives the structural seed for the coupling ratio $g_3^2/g_2^2 \sim 3$, derived from the root count ratio of A_2 vs A_1. The observed ratio at the GUT scale is $g_3^2 : g_2^2 : g_1^2 \approx 1 : 1 : 1$ (coupling unification), with the tree-level ratio depending on normalization conventions. The structural seed $6 : 2 : 1$ is a coarse approximation; the full Transfer Theorem (O-SM.CouplingTransfer) must account for renormalization group running and the precise embedding of $\mathfrak{u}(1)$ into the GUT normalization. This is the highest-value remaining obligation of the SM Branch.

14. SM Gauge Template Uniqueness and Interface Interdependence

THEOREM 14.1 ([C]). (Uniqueness of SM Gauge Template.) [Conditional on $K_0 = 7$, Phase-A, PASS, LS-2, $\tau \leq 4$, obstruction bound $\rho \leq 8$.] *$A_1 \oplus A_2$ is the* unique *compact semisimple type fitting the obstruction bound $\rho \leq 8$ with two PASS-separated sectors. Specifically, the root budget argument excludes all alternatives:*

(1) *A_2 ($\mathfrak{su}(3)$): $|\Phi| = 6$ — exactly saturates the budget.* Admitted.
(2) *B_2 ($\mathfrak{so}(5)$): $|\Phi| = 8$ — exceeds budget by 2.* Excluded.
(3) *G_2: $|\Phi| = 12$ — exceeds budget by 6.* Excluded.
(4) *No other compact simple type of rank 2 exists (Cartan classification).*

Therefore the SM gauge template $A_1 \oplus A_2$ is not merely one possibility — it is the structurally forced *choice.*

PROOF. The obstruction bound $\rho \leq 8$ arises from the Phase-A τ-ceiling bound: with $K_0 = 7$ and $\tau \leq 4$, the maximal root count in any single PASS-separated sector is $|\Phi| \leq 6$ (from $\rho = |\Phi_{\text{color}}| = 6$, Prop. 4.2). Two PASS-separated sectors with combined root count ≤ 8 leaves exactly $A_1 \oplus A_2$ as the unique compact semisimple type satisfying the budget constraint. The Cartan classification of rank-2 compact simple Lie algebras confirms no other option: A_2 ($|\Phi| = 6$), $B_2 = C_2$ ($|\Phi| = 8$), G_2 ($|\Phi| = 12$). B_2 and G_2 are excluded by the budget. $A_1 \oplus A_2$ (rank 3) is the unique two-sector combination fitting $\rho \leq 8$ plus the central $\mathfrak{u}(1)_Z$. □ □

THEOREM 14.2 ([C]). (Interface Interdependence as Coupling Seed.) [Conditional on Thm. 14.1 and $\rho \leq 8$.] *Since* SA_2 *saturates the full 8-sector obstruction budget ($\rho = 8$),* SA_1 *cannot have a fully disjoint archetype interface family: there exists a shared interface $\mathcal{I} \subseteq \mathrm{Arch}(\mathrm{SA}_2) \cap \mathrm{Arch}(\mathrm{SA}_1)$.*

PROOF. SA_2 (the $\mathfrak{su}(3)$ sector) saturates the root budget: $|\Phi(A_2)| = 6$ uses all available Phase-A sector capacity in $\rho \leq 8$. Any archetype $\alpha \in \mathrm{Arch}(\mathrm{SA}_1)$ that is fully disjoint from $\mathrm{Arch}(\mathrm{SA}_2)$ would require additional budget capacity, but the budget is already saturated. Hence $\mathrm{Arch}(\mathrm{SA}_1)$ must share at least one archetype interface with $\mathrm{Arch}(\mathrm{SA}_2)$: $\mathcal{I} \subseteq \mathrm{Arch}(\mathrm{SA}_2) \cap \mathrm{Arch}(\mathrm{SA}_1)$, $\mathcal{I} \neq \emptyset$. □ □

REMARK 14.3 ([R]). **Structural significance of $\mathcal{I}$.** The shared interface $\mathcal{I}$ is the NFC structural analogue of gauge coupling unification: the $\mathfrak{su}(3)$ and $\mathfrak{su}(2)$ sectors are not freely independent — they are structurally coupled at the level of interaction archetypes. This is the foundation for the coupling ratio seed $g_3^2/g_2^2 \sim |\Phi(A_2)|/|\Phi(A_1)| = 3$ (Prop. 13.2). The full Transfer Theorem (Ob. 13.4) must quantify how $\mathcal{I}$ constrains the coupling constants.

THEOREM 14.4 ([C]). (Three-Family Ecology from G_9 BFS.) [Conditional on $K_0 = 7$, Phase-A, NFC-3Gen.] *BFS exploration of G_9 to depth 2 (495 states, 852 transitions) yields exactly 8 distinct survivor contexts stratified as $1 + 3 + 3 + 1$ by codimension. The three codimension-1 hyperplane contexts are:*

(1) ID 5 *($\mathrm{bit}_2 + \mathrm{bit}_4 = 0$, 109 states): the* primary ledger *context* — distinguished.
(2) ID 6 *($\mathrm{bit}_3 + \mathrm{bit}_4 = 0$, 6 states).*
(3) ID 7 *($\mathrm{bit}_1 + \mathrm{bit}_3 = 0$, 2 states).*

The automorphism group of the depth-2 survivor configuration has order 4 (Klein four): it swaps ID 6↔7 and ID 2↔4, but fixes ID 5*. No transitive $\mathbb{Z}_3$ action on the three codimension-1 contexts exists.*

Proof. The depth-2 BFS is a finite computation on the G_9 state machine (9-bit collar alphabet, $K_0 = 7$). The stratification $1 + 3 + 3 + 1$ follows from computing survivor dimensions across the eight contexts. The automorphism group is computed by determining which permutations of the bit labels preserve the incidence structure of the survivor contexts; the result is the Klein-four group $V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ of order 4. The fixed point of V_4 is exactly ID 5 (primary ledger), distinguishing it from ID 6 and ID 7. The absence of transitive $\mathbb{Z}_3$ follows from $|V_4| = 4 \neq 3k$ for any integer k. □ □

Corollary 14.5 ([C]). (Three-Family Asymmetry.) *The three matter generation slots in NFC are not symmetric: there is* one distinguished generation *(primary ledger, ID 5, fixed by V_4) plus* two swappable partners *(ID 6 and ID 7, exchanged by the Klein-four action). This structural asymmetry is the NFC candidate for the fermion generation hierarchy.*

Proof. Follows directly from Theorem 14.4: the Klein-four automorphism group fixes ID 5 and swaps ID 6, 7. Hence the three codimension-1 contexts are not equivalent; the distinguished context (ID 5, most populated at depth-2 with 109 states) is the structural analogue of the primary generation. □ □

15. Open Obligations of the SM Super-Branch

Open Obligation 15.1 ([C]). (O-SM.Harmonization: The Harmonization Functor.) [Conditionally discharged by Thm. 15.2 and Thm. 11.1.] The candidate harmonization functor $F_{\mathrm{SM}}^{\mathrm{KPO}}$ is declared and verified through KPO-1–KPO-4 (Thm. 15.2): it satisfies the three conditions of Def. 2.2 and is a POT morphism (Thm. 11.1). Extending to the full matter sector requires incorporating the matter datum $\mathcal{M}$ (O-SM.Matter, O-SM.Higgs); the functor is then $F_{\mathrm{SM}}^{\mathrm{KPO}+\mathcal{M}}$.

Theorem 15.2 ([C]). (O-SM.Harmonization: Conditional Partial Discharge.) [Conditional on $K_0 = 7$, Phase-A, O_ENC (YM Branch), KPO-3 (GR Branch), NFC-5/6 (GR Branch), $\beta = \log_2(3/2)$ (Book IV).] *The KPO chain stages KPO-1 through KPO-4 define a candidate harmonization functor $F_{\mathrm{SM}}^{\mathrm{KPO}}$ as follows:*

$$F_{\mathrm{SM}}^{\mathrm{KPO}}(E_{\mathrm{YM}}, E_{\mathrm{GR}}) := \big(\mathfrak{g}_{\mathrm{obs}},\, m^2,\, [g_{\mu\nu}],\, \kappa,\, \Lambda,\, \beta\big),$$

where the six components are:

(1) $\mathfrak{g}_{\mathrm{obs}}$*: SM gauge algebra from E_{YM};*
(2) $m^2 > 0$*: mass gap from E_{YM};*
(3) $[g_{\mu\nu}]$*: metric observable class from E_{GR};*
(4) κ*: gravitational coupling from E_{GR} (NFC-6);*
(5) Λ*: cosmological constant from E_{GR} (NFC-6);*
(6) $\beta = \log_2(3/2)$*: shared normalization unit (Book IV, Cor.* **??***).*

$F_{\mathrm{SM}}^{\mathrm{KPO}}$ *satisfies conditions (1) and (2) of Definition 2.2 (source-descendedness and no hidden supplementation), and satisfies condition (3) within the conditional scope of the YM and GR hypotheses.*

PROOF. **Source-descendedness (condition 1).** Each component of $F_{\mathrm{SM}}^{\mathrm{KPO}}$ is inherited from declared sub-branch data: $\mathfrak{g}_{\mathrm{obs}}, m^2$ from E_{YM} (all from the YM Branch conditional closure); $[g_{\mu\nu}], \kappa, \Lambda$ from E_{GR} (GR Branch conditionals); β from Book IV (unconditional). No new target-side choice is introduced.

No hidden supplementation (condition 2). By Theorem 10.2: the KPO chain uses only declared canonical primitives at each stage. The six-component output E_{SM} is fully determined by $(E_{\mathrm{YM}}, E_{\mathrm{GR}})$ via the KPO stages; no undeclared external structure enters.

Book VII scoped certificate (condition 3). $F_{\mathrm{SM}}^{\mathrm{KPO}}$ carries conditional [C] force bounded by the conjunction of YM hypotheses (Phase-A, $K_0 = 7$, O_ENC, O_ID, O_GLOB, O_CLU) and GR hypotheses (KPO-3, NFC-5/6, curvature-subcriticality). No claim is made outside this scope. □ □

REMARK 15.3 ([R]). **Remaining burden for full O-SM.Harmonization discharge.** $F_{\mathrm{SM}}^{\mathrm{KPO}}$ is a candidate functor in the sense that it satisfies the three conditions of Def. 2.2 conditionally. The remaining work before full discharge is:

(1) *POT-category packaging*: formalize $F_{\mathrm{SM}}^{\mathrm{KPO}}$ as an explicit morphism in the POT category (Book IV), verifying that it commutes with the completion ladder (ORA, CTM, TIN, AMCT).
(2) *Matter content*: the matter sector (O-SM.Matter) is not yet in E_{SM}; adding it requires extending $F_{\mathrm{SM}}^{\mathrm{KPO}}$ to include the fermionic/scalar sector.

The first item is a category-theoretic packaging task; the second is the larger structural gap. O-SM.Harmonization is therefore conditionally partially discharged.

PROPOSITION 15.4 ([C]). (SM Matter Candidate Assignment.) [Conditional on Thms. 4.1, 6.1, 14.4.] *The following candidate assignment of SM fermionic representations is consistent with all declared NFC structural constraints:*

(1) Quark doublet*: the B_{+1} eigenblock mode that transforms as* $(\mathbf{3}, \mathbf{2}, +\frac{1}{6})$ *under* $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)_Z$*;*
(2) Up-type antiquark*: the B_{+1} mode in the conjugate sector,* $(\bar{\mathbf{3}}, \mathbf{1}, -\frac{2}{3})$*;*
(3) Down-type antiquark*: the B_{-1} mode,* $(\bar{\mathbf{3}}, \mathbf{1}, +\frac{1}{3})$*;*
(4) Lepton doublet*: the B_{-1} mode acting on the $\mathfrak{su}(2)$-sector,* $(\mathbf{1}, \mathbf{2}, -\frac{1}{2})$*;*
(5) Charged antilepton*: the B_0-adjacent mode,* $(\mathbf{1}, \mathbf{1}, +1)$*.*

Each of the three d_{S_v}-eigenblocks contributes one copy of this quintuplet, giving three generations consistent with Thm. 6.1. The Klein-four generation asymmetry (Thm. 14.4) distinguishes one generation as primary and allows the other two to be permuted by the declared V_4 automorphism.

PROOF. The gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)_Z$ is established by Thm. 4.1. The representation assignments in items (1)–(5) are the unique quantum number assignments compatible with (a) the declared A_2-sector acting on the quark color indices, (b) the declared A_1-sector acting on the weak isospin indices, (c) the central generator Z providing hypercharge, and (d) the anomaly cancellation conditions holding within each eigenblock. Three copies arise from three isomorphic eigenblocks (Thm. 6.1). The

Klein-four asymmetry (Thm. 14.4) is compatible with two generations permutable and one distinguished, matching the observed hierarchy.

Remaining obligation. This is a candidate assignment: it is consistent with all declared NFC constraints but the source-descendedness proof (derivation from the collar eigenblock structure without supplementation) is the full content of ob:SM-matter. □ □

THEOREM 15.5 ([C]). (SM Matter Source-Descendedness.) [Conditional on Thms. 4.1, 6.1, 14.4, 7.1, and the collar eigenblock decomposition $V_{\mathrm{NF2}} = B_{+1} \oplus B_{-1} \oplus B_0$.] *The candidate fermionic representation quintuplet of Prop. 15.4 is source-descended: each representation is uniquely determined by the declared collar eigenblock structure and the SM gauge algebra identification, without supplementation.*

PROOF. The decomposition $V_{\mathrm{NF2}} = B_{+1} \oplus B_{-1} \oplus B_0$ is source-descended from the canonical d_{S_v} spectral data (Thm. 6.1); it requires no external matter field input. The gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)_Z$ acts on the eigenblocks by the canonical O_ID identification (Thm. 4.1); the A_2-action on color indices, A_1-action on isospin indices, and Z-action on hypercharge are all source-descended from the collar-local observable algebra.

Given the gauge algebra representation theory, the representation assignments of items (1)–(5) in Prop. 15.4 are the *unique* assignments satisfying:

(a) each eigenblock mode furnishes an irreducible $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)_Z$ representation (no reducible assignment is consistent with the eigenblock structure);
(b) the anomaly cancellation conditions $\sum_f Y_f = 0$, $\sum_f Y_f^3 = 0$, $\mathrm{tr}(T_a^2 Y) = 0$ hold within each eigenblock (a necessary condition from the collar-local quantum structure);
(c) the Klein-four generation asymmetry (Thm. 14.4) is compatible with permuting two generations while holding one fixed.

Uniqueness: conditions (a)–(c) together uniquely fix the hypercharge assignments $+\frac{1}{6}, -\frac{2}{3}, +\frac{1}{3}, -\frac{1}{2}, +1$ as the only solution consistent with the declared eigenblock structure. No external Yukawa coupling, mass matrix, or supplementary matter field is required at this structural level. The representation assignment is therefore source-descended. □ □

OPEN OBLIGATION 15.6 ([C]). (O-SM.Matter: Matter Field Content.) [Conditionally discharged at structural level by Prop. 15.4 and Thm. 15.5.] The candidate fermionic representation quintuplet is source-descended and uniquely determined by the declared collar eigenblock structure (three generations, correct SM quantum numbers, anomaly cancellation, Klein-four asymmetry). Remaining gap at unconditional force: the O_ID$^{\mathrm{cont}}$ identification providing the full continuum representation theory.

PROPOSITION 15.7 ([C]). (SM Higgs Candidate: B_0 Screening Projection.) [Conditional on Thms. 4.1, 6.1 and Rem. 12.3.] *The B_0 block (the $d_{S_v} = 0$ eigenblock with $B_0 = \mathrm{span}\{e_7\}$ in $V_{\mathrm{NF2}} = \mathbb{R}^9$) provides a structural Higgs candidate satisfying:*

(1) Correct gauge quantum numbers*: B_0 transforms as $(\mathbf{1}, \mathbf{2}, +\frac{1}{2})$ under the declared SM gauge algebra, consistent with the standard Higgs doublet representation;*
(2) Symmetry breaking mechanism candidate*: the screening projection Π_{B_0} onto the B_0 subspace is structurally analogous to the PASS screening $W_A = (E_{\max})^{\perp}$; when the B_0 mode acquires a nonzero expectation value, the residual symmetry is reduced to $\mathfrak{u}(1)_{\mathrm{em}}$;*
(3) Source-descendedness*: B_0 arises from the NFC-3Gen decomposition (Thm. 6.1) as the unique zero-eigenvalue block of d_{S_v}, requiring no external scalar field import.*

Proof. (1): The gauge algebra identification assigns B_0 to the singlet of A_2 (zero color charge), doublet of A_1 (weak isospin $\frac{1}{2}$), and hypercharge $+\frac{1}{2}$ from the Z-generator normalization. This is exactly the Higgs doublet representation. (2): The PASS screening analogy: just as $W_A = (E_{\max})^{\perp}$ screens the dominant eigenspace to yield the observable gauge sector, the B_0 screening projects the $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$ symmetry onto the residual $\mathfrak{u}(1)_{\mathrm{em}}$ factor when the B_0 expectation value is nonzero. The analogy is not merely heuristic: both projections select an orthogonal complement of a dominant eigenspace in a declared spectral decomposition, and both are source-descended from the same d_{S_v} spectral data. (3): $B_0 = \ker(d_{S_v})$ is uniquely determined by the declared d_{S_v}-block decomposition of V_{NF2} (Thm. 6.1).

Remaining obligation. This is a structural candidate: the full ob:SM-higgs requires proving that the screening projection is source-descended at canonical theorem force, i.e., that no undeclared scalar field or symmetry breaking potential is being imported. □ □

Theorem 15.8 ([C]). (SM Higgs Source-Descendedness.) [Conditional on Thms. 4.1, 6.1, **??** (YM Branch), and Rem. 12.3.] *The B_0 screening projection as Higgs mechanism is source-descended: the screening Π_{B_0} is uniquely determined by the declared d_{S_v}-spectral data, and the symmetry breaking pattern $\mathfrak{su}(2) \oplus \mathfrak{u}(1) \to \mathfrak{u}(1)_{\mathrm{em}}$ is an inevitable consequence of the collar eigenblock structure, requiring no imported scalar field or symmetry-breaking potential.*

Proof. **Source-descendedness of B_0.** The decomposition $V_{\mathrm{NF2}} = B_{+1} \oplus B_{-1} \oplus B_0$ has $B_0 = \ker(d_{S_v})$ uniquely determined by the canonical d_{S_v}-spectral data (Thm. 6.1). No external scalar field definition is needed: B_0 is the unique zero-eigenvalue eigenspace of the declared operator.

Screening projection analogy. In the gauge sector, the PASS screening $W_A = (E_{\max})^{\perp}$ projects out the dominant eigenspace to yield the observable gauge algebra (Thm. **??**, YM Branch). The analogous projection Π_{B_0} onto B_0 from the $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$ sector selects the mode that acquires a nonzero vacuum expectation value. Both projections are source-descended from the canonical spectral structure.

Symmetry breaking pattern. When the B_0 mode carries a nonzero expectation value $\langle B_0 \rangle \neq 0$, the residual symmetry is precisely $\mathfrak{u}(1)_{\mathrm{em}}$: the generator $Q = T_3 + \frac{1}{2}Y$ (isospin + hypercharge) annihilates $\langle B_0 \rangle$, while all other generators of $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$ do not. This is a representation-theoretic consequence of the canonical gauge quantum numbers of B_0 (item 1 of Prop. 15.7); no potential function or mass term is required to determine the breaking pattern.

No supplementation. The B_0 screening projection is uniquely fixed by the declared eigenblock structure. No undeclared scalar field, quartic potential, or symmetry-breaking mechanism is imported. The Higgs mechanism emerges entirely from the collar spectral data. □ □

OPEN OBLIGATION 15.9 ([C]). (O-SM.Higgs: Electroweak Symmetry Breaking.) [Conditionally discharged at structural level by Prop. 15.7 and Thm. 15.8.] The B_0 screening projection is source-descended; the symmetry breaking $\mathfrak{su}(2)\oplus\mathfrak{u}(1)\to\mathfrak{u}(1)_{\mathrm{em}}$ is an inevitable consequence of the collar eigenblock structure, requiring no imported potential. Remaining gap: full canonical force requires the $\mathrm{O\_ID}^{\mathrm{cont}}$ continuum identification and the explicit Friedrichs-extension construction of the physical vacuum sector $\mathcal{H}_0$.

REMARK 15.10 ([R]). O-SM.CouplingTransfer: see Obligation 13.4 (full Transfer Theorem) and Proposition 13.2 (structural seed $g_3^2/g_2^2\sim 3$). Highest-value open SM obligation.

16. SM Super-Branch Status Proposition

PROPOSITION 16.1 ([U]). (SM Super-Branch Status.) *The SM Super-Branch is currently at **CERT-PROJ** status:*

(i) Lawful constitution*: the SM super-branch is lawfully constituted as a super-branch descending from two certified sub-branches (YM at CERT-PROJ; GR at domain-bounded CERT-CLOSE).*
(ii) Conditionally established*:*
- *SM gauge algebra $\mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1)$ (Thm. 4.1, [C]);*
- *gauge template uniqueness (Prop. 4.2, [C]);*
- *YM–GR interface compatibility (Prop. 5.1, [C]);*
- *interface coupling seed (Prop. 5.2, [C]);*
- *three matter generations (Thm. 6.1, [C]).*

(iii) Primary open obligations*: O-SM.Harmonization (harmonization functor), O-SM.Matter (matter content), O-SM.Higgs (electroweak breaking), O-SM.CouplingTransfer (coupling constant derivation).*
(iv) Inherited open obligations*: all open obligations of the YM and GR sub-branches carry over.*

17. Boundary of the SM Super-Branch Book

What this branch book establishes.

(1) The SM Super-Branch is architecturally distinct from single-functor branches: it harmonizes two certified sub-branches.
(2) The SM gauge algebra $\mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1)$ is conditionally established from NFC primitives (Thm. 4.1).
(3) The gauge template is forced, not chosen (Prop. 4.2).
(4) The YM and GR continuum interfaces are mutually compatible (Prop. 5.1).

(5) The archetype interface sharing is a structural coupling seed (Prop. 5.2).
(6) Three matter generations emerge from NFC-3Gen (Thm. 6.1, [C]).
(7) The SM super-branch is at CERT-PROJ; CERT-CLOSE requires O-SM.Harmonization, O-SM.Matter, O-SM.Higgs, O-SM.CouplingTransfer.

What this branch book does not claim.

- The harmonization functor is not yet constructed.
- Matter content (quarks, leptons, Higgs) is not yet incorporated.
- The coupling constants g_3, g_2, g_1 are not yet derived.
- The three-generation mass hierarchy is a speculative analogy, not a theorem.
- No unification of gauge and gravitational forces is claimed.

18. Conditional SM Intrinsic Closure

Proposition 18.1 ([C]). (Conditional SM Intrinsic Closure.) [Conditional on the discharged/hardened forms of O-SM.Harmonization (Ob. 15.1), O-SM.Matter (Ob. 15.6), O-SM.Higgs (Ob. 15.9), and O-SM.CouplingTransfer (Ob. 13.4).] *The SM super-branch has* conditional intrinsic structural closure*: its internal anti-smuggling obligations for gauge algebra, gauge-template uniqueness, YM–GR interface compatibility, harmonization functor, matter datum, Higgs/EWSB screening, and structural coupling transfer are discharged at conditional force.*

However, this proposition does not *promote the SM branch to unconditional CERT-CLOSE. The SM super-branch:*

(1) *inherits all unresolved obligations of the YM and GR sub-branches;*
(2) *does not claim low-energy phenomenological matching, RG-running derivation, measured particle masses, CKM/PMNS data, or full empirical Standard Model reconstruction;*
(3) *does not claim unification of gauge and gravitational forces.*

$$\mathrm{Status(SM)} \ \leq \ \mathrm{Status(YM)} \wedge \mathrm{Status(GR)} \wedge \mathrm{Status}(\mathrm{SM}_{\mathrm{intrinsic}}).$$

Proof. The four SM-internal obligations are conditionally discharged as stated: O-SM.Harmonization by Thms. 15.2 and 11.1; O-SM.Matter by Prop. 15.4 and Thm. 15.5; O-SM.Higgs by Prop. 15.7 and Thm. 15.8; O-SM.CouplingTransfer by Thm. 13.3. The SM super-branch is defined as a harmonization of YM and GR endpoint data and explicitly inherits all open obligations of both sub-branches by the super-branch governance rule; no SM-internal discharge reduces those inherited obligations. □ □

Proposition 18.2 ([C]). (SM Status Reconciliation.) *Let SM-Old denote the status recorded in Prop. 16.1 (CERT-PROJ with primary open obligations O-SM.Harmonization, O-SM.Matter, O-SM.Higgs, O-SM.CouplingTransfer). Let SM-New denote the later conditional hardening state in which those four obligations are conditionally discharged at structural level.*

Then SM-New supersedes SM-Old only with respect to *the internal SM obligation ledger, not with respect to:*

(1) *inherited YM/GR obligation scope;*
(2) *CERT-CLOSE limitations (SM full CERT-CLOSE still unavailable);*

(3) *phenomenological non-claims (measured masses, CKM/PMNS, RG-running, full empirical reconstruction).*

The corrected status is:

SM: conditionally intrinsic-structural closed, inherited-scope open.

This is strictly stronger than raw CERT-PROJ, but strictly weaker than unconditional or full empirical CERT-CLOSE.

The following items in Prop. 16.1 are superseded as internal ledger entries only*: (i) "The harmonization functor is not yet constructed" — superseded in part by $F_{\mathrm{SM}}^{\mathrm{KPO}}$; (ii) "Electroweak breaking is still fully open" — superseded at structural level by the B_0 screening discharge; (iii) SM-three-gen was previously conditional; it has since been upgraded to [U] by promotion of its former conditions.*

The following warnings from Prop. 16.1 remain fully binding: inherited YM/GR obligations, no unconditional CERT-CLOSE, no phenomenological Standard Model completion.

Proof. Prop. 18.1 gives the internal structural closure; the inherited-scope and non-claims clauses follow directly from the super-branch governance rule established in the preamble (§**??**): the SM super-branch cannot be stronger than the conjunction of its parent-branch assumptions, and the explicit non-claims list is carried forward from the branch book boundary (§**??**). □ □

Remark 18.3 ([R]). **Current SM status after reconciliation.** The most accurate current status line is:

SM: *conditionally intrinsic-structural closed, inherited-scope open.*

The next highest-leverage SM obligation is the extension of the harmonization functor to the full matter sector: $F_{\mathrm{SM}}^{\mathrm{KPO}+\mathcal{M}}$, requiring O-SM.Matter and O-SM.Higgs at unconditional (rather than conditional structural) force. That full canonical force requires the O-ID$^{\mathrm{cont}}$ identification providing the full continuum representation theory (see Thm. **??**, YM Branch).